Maxwell Goldenson

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SUMMARY

Robotics and systems engineer pursuing dual degrees in Mechanical and Electrical Engineering ~~at CU Boulder~~. Experienced in designing, integrating, and debugging real-world robotic and embedded systems that combine precision mechanisms, embedded electronics, sensors, control, and AI software. Strong background in prototyping, reliability testing, and system-level problem solving.

EXPERIENCE

Brewbird, Smart Coffee Machine — **San Francisco, CA**
Associate Mechanical Engineer — May 2024 – Present

- Designed and built the AutoCycler, an unattended endurance-test fixture for Brewbird machines: stepper auger doser, servo door actuator, capacitive brew trigger, and color sensors reading the status lights, on two ESP32s and a Raspberry Pi.
- Wrote its full software stack: ESP32 firmware and serial protocol, a Python touchscreen app that closes the loop on the status lights and pushes phone alerts, and an OTA launcher that updates the app and firmware fleet-wide from GitHub.
- Debugged and resolved several mission critical cross-functional issues, reducing service events and achieving an annual service time reduction of >2500h.
- Redesigned and validated the Pod Opener Mechanism, extending reliable operation by 500+ cups.
- Existing solutions lacked a balance of size and accuracy, developed a novel linear encoder from theory to prototype achieving +- 10um precision, improving grind size accuracy, repeatability, and ultimately taste.

Switchback, AI Assistant over SMS & Satellite (switchbackconnect.com) — **Boulder, CO**
Founder & Software Engineer — Aug 2026 – Present

- Founded Switchback, an AI assistant hikers and climbers text over SMS or satellite (iPhone, Garmin inReach); a Claude agent with live weather, terrain, and web-search tools answers backcountry questions in ≤140-character messages.
- Architected and shipped a serverless Python backend (AWS Lambda, DynamoDB, Twilio, Stripe, Claude API) from scope to production in 6 days at ~$0 fixed monthly cost, with 47 automated tests and three-tier Stripe subscription billing.
- Built the agent loop with two-tier model routing (Haiku triage, Sonnet escalation), tool use, a code-enforced GSM-7 pipeline capping replies at 2 × 140 chars, and an SOS-first safety rail; validated on real satellite hardware.

SIGNIFICANT PROJECTS

San Francisco & Boulder

- **Tools-For-AI, AI Hardware-in-the-Loop Debug Rig (2026):** Built a Raspberry Pi bench node that gives an AI agent real instruments (logic-analyzer capture/decode, signal measurement, ESP32 compile/flash, serial) as 24 MCP tools. The agent writes firmware, flashes it, measures, and revises until machine-checked pass/fail criteria are met, so every conclusion rests on a real measurement, not the model's own claims. Full Python stack: instrument back end, Claude agent loop with an injection boundary for untrusted serial and compiler text, browser Workbench/Assistant app, mock mode, 75 tests.
- **ESP32-Based High Speed Data Logger (2025):** Developed a custom ESP32 system for high-frequency, long-term data acquisition and debugging, using efficient memory buffering to log over 10k samples/second continuously for days.
- **Linear Capacitive Encoder (2024):** Designed, prototyped, and validated a high-accuracy linear position sensor using plate capacitance, achieving +-10 µm resolution for <$5.
- **High-Speed 3D Printers (2022–2023):** Engineered a delta-style printer that benchmarked faster than comparable commercial printers, then modeled a record-speed CoreXY concept in full CAD. (SOLIDWORKS, Onshape)
- **Food-Waste-Reducing Smart Tongs (2023):** Led a team to build ESP32 portion-control tongs; won a sustainability award.

EDUCATION

University of Colorado Boulder

- Bachelor of Science — Mechanical Engineering (BSME)
- Bachelor of Science — Electrical Engineering (BSEE)
- Junior standing GPA: 3.68

SKILLS

SOLIDWORKS • OnShape • Quick-Learner • Leadership • Schematic Capture • C • Arduino-C • Python • LLM Agents • React • Verilog • Rapid Prototyping • Manufacturing • Matlab • SiMetrix (Spice) • 3DP & CNC Operation • Injection Molding